

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Chemistry

2. Subject Name: Chemistry of Main Group Elements: Ring, Chains, Clusters L-T-P: 3-0-0 Credits: 3

3. Pre-requisites: Inorganic Chemistry II

4. Syllabus and reference books:

Syllabus:

Rings, chains, clusters and polymers of s, p, d-block elements. Boranes, Carboranes, Metalloboranes, Siloxanes, types of silicates, Phosphazenes, Borazine, P-N, P-O, S-N rings, polymers, and other rings systems of Al, P, S and their properties. Wades/Mingos rules for both main group and transition metal clusters for their geometries and electronic structures. Principle of isolobal analogy for main group and transition metal fragments. Transition to the solid state.

Reference Books:

- (1) Inorganic Chemistry-Principles of Structure and Reactivity. J. E. Huheey, E. A. Keiter, R. L. Keiter, 4th Edn. Harper-Collins, NY, 1993.
- (2) Inorganic Rings and Polymers of *p*-block elements: Tristram Chivers and Ian Manners, RSC publishing, 2009.
- (3) Inorganic Chemistry by Shriver and Atkins. International Student edition, fourth edition, Oxford University press 2008.
- (4) Molecular Clusters Thomas P. Fehlner, Jean-Francois Halet and Jean-Yves Sailer, Cambridge University Press, 2008
- (5) Advanced Inorganic Chemistry by Cotton and Wilkinson, 6th edition, Wiley, 1999.

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Boranes, Carboranes, Metalloboranes,	10
2.	Siloxanes, types of silicates, Phosphazenes, Borazine, P-N, P-O, S-N rings, polymers, and other rings systems of Al, P, S and their properties.	10
3.	Wades/Mingos rules for both main group and transition metal clusters for their geometries and electronic structures. Principle of isolobal analogy for main group and transition metal fragments. Transition to the solid state.	12
Total number of hours		32